SYSTEM USER MANUAL

Document Title:	Wholesale ERP System User Manual
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Reference:	ERP-UM-v1.0
Target Audience:	System Operators, Management, and Corporate Clients

1. ERP System Modules Overview

The Wholesale ERP is a fully integrated, centralized enterprise resource planning platform designed to unify manufacturing, inventory asset management, procurement workflows, client dispatches, financial ledgers, and workforce compliance tracking. Information propagates across database layers automatically to protect absolute transactional consistency.

A. Catalog & Parts Management

Serves as the root directory for tracking physical architecture. Houses items across distinct lifecycle categories:

- **Raw Materials:** Raw commodities or consumable bulk components entering standard production.
- **Semi-Finished Goods:** Processed intermediate stocks currently undergoing factory transformations before outer packaging.
- **Finished Goods:** Completed product units ready for commercial sales and bulk distribution networks.
- **Packaging Materials:** Direct storage physical boxes, inner wrappers, security seals, and distribution pallets.

B. Procurement & Receiving (GRN)

Bridges operations with active external supplier contracts.

- **Purchase Orders (PO):** Legally binding procurement records dictating agreed unit costs, strict counts, line details, and delivery timelines to specific destination warehouses.
- **Goods Receipt Notes (GRN):** Tracks physical items through real-time operational interfaces. Warehouse workers process lines, logging damaged items directly into automated fields. If an active recipe structure exists, the interface delivers a dynamic live processing yield estimation instantly.

C. Manufacturing & MES (Manufacturing Execution System)

Facilitates full tracking of shop-floor executions and processes.

- **Process Templates:** Structured recipe workflows defining strict operations sequences (e.g., cleaning, initial slicing, dehydration, milling, and box sealing).

- **Production Batches:** Monitors real-time work runs. Seamlessly records operational shifts (day/night breakdowns), active labor assignments, drying mechanics, facility utility resource draws (measured firewood weights and net kilowatt-hour electricity metrics), and weight yields.
- **Quality Control (QC):** Completed inventory runs automatically queue into rigorous chemical and physical testing workflows. Stock units transition to trade-ready inventory layers exclusively following official Quality Officer validation updates.

D. Sales, Invoicing & Logistics

Oversees core commercial downstream logistics.

- **Sales Orders (SO):** Tracks corporate client purchases, special pricing matrix assignments, and explicit fulfillment nodes.
- **Invoices:** Generates formalized tax invoices natively structured around regional tax calculations (defaulting to 18% VAT adjustments).
- **Logistics & Gate Passes:** Automated packaging flows generate specialized, secure transport clearances. These detail assigned transport asset configurations, operator identifiers, and dynamic package breakdown arrays to validate compound egress. Fleet systems log ongoing fuel draws, mileage, and task milestones.

E. Petty Cash & Bank Management

Maintains high-integrity corporate monetary tracking records.

- **Petty Cash:** Governs baseline facility cash boxes through multi-level validation steps.
- **Bank Accounts:** Reconciles official operational banking infrastructure against arriving customer remittances and outgoing manufacturing liabilities.
- **Cheque Ledger:** Tracks long-dated payment clearances and operational verification states.

F. HR & Payroll Module

Coordinates corporate organizational staff registries and employee benefits.

- **Employee Directory:** Stores professional positions, regional divisions, identity indicators, and structured shifts.
- **Attendance Logs:** Syncs clock terminal interactions with predefined roster requirements.
- **Payrolls:** Produces formal structural compensation summaries. Computes baseline salaries, active overtime parameters, bonuses, financial deductions, and explicit individual payout breakdowns automatically.

2. Step-by-Step Operational Guides

How to Create a Part / Product

1. Authenticate securely within the platform holding **Admin** or **Manager** structural access permissions.
2. Access the primary navigation block, enter the **Products** module, and activate the **Add Product** terminal button.

3. Within the *Basic Info* subsection, observe that the **Product Code** is locked and initialized to *"Auto-generated after category selection"*.
4. Define the required organizational classification via the **Category** dropdown (such as Mechanical or Raw Materials).
5. The underlying system engine instantly queries the active indexing registry, produces a unique Julian date tracking code, and populates the **Product Code** box. The parameter remains strictly write-locked to maintain database structural safety.
6. Provide the clear **Product Name**, **Short Name**, **Type** (e.g., trading, manufactured), and appropriate **Unit of Measure**.
7. Input relevant context inside companion areas (*Pricing & Tax*, *Stock & Packaging*, *Sales Config*) and commit the changes via **Save Product**.

How to Receive Inbound Raw Materials (GRN)

1. Navigate to the **Purchase Orders** dashboard and access an open, unfulfilled line item entry.
2. Select the **Receive Goods** element to reveal the contextual GRN modal window.
3. For each scheduled line, log the net **Received Qty** and **Rejected Qty** following initial quality inspections. The application computes the net valid weight automatically:

$$\text{Accepted Qty} = \text{Received Qty} - \text{Rejected Qty}$$

1. If an active conversion recipe governs the target asset, the modal window displays a **Live Yield Forecast** section, outlining expected finished production outputs from accepted weights.
2. Log essential metadata: **Supplier Invoice #**, **Supplier Delivery Note #**, **Vehicle Number**, **Driver Name**, and explicit **Expiry Dates** for perishable goods.
3. Commit the form using **Confirm Receipt** to instantly commit quantities to active warehouse inventory records.

How to Log and Complete a Production Batch

1. Access the manufacturing workspace, navigate to **Batches**, and click **Log New Batch**.
2. Designate the target **Product** to be manufactured, the originating raw material **Supplier**, and the structural **Process Template**.
3. Log structural workforce metrics: **Staff Headcount** (split by Day/Night shift rosters) along with total **Input Weights (Kg)**.
4. Log environmental resource utility indicators: active **Drying Hours**, total **Firewood Consumed (Kg)**, and measured **Electricity Units (kWh)**.
5. Upon process termination, record the net **Output Weights (Kg)** and commit the file.
6. The system alters the run status to **"Completed"**, immediately placing the batch in the **QC Checks** verification pipeline.

7. An authorized **Quality Officer** verifies analytical lab parameters and selects **Approve Batch** to advance clean inventory into commercial channels.

3. Auto-Calculations Engine

The platform eliminates human mathematical error via localized backend calculations engines across core operations modules.

A. Bill of Materials (BOM) Costing

When compiling structural formulation matrix definitions (BOM), standard unit asset cost calculations combine ingredient raw costs, systematic yield allowances, manual labor investments, and generic production markups:

$$Total\ Cost = \Sigma\ (\ Component\ Qty \times (\ 1 + Wastage\% / 100\ ) \times Unit\ Cost\ ) + Labor\ Cost + Overhead\ Cost$$

- **Component Qty:** The initial benchmark material weight or count constraint.
- **Wastage %:** Factor adjustments mapping normal item volume degradation throughout processing.
- **Labor Cost:** Total cumulative work hours multiplied by specific regional processing pay-tier structures.
- **Overhead Cost:** An automated proportional calculation mapping asset costs to handle continuous plant machinery expenditures.

B. Production Batch Efficiency

Following batch completion, the tracking engine grades raw conversion metrics to measure output performance:

$$Efficiency\% = (\ Output\ Weight\ (Total) / Input\ Weight\ (Total)\ ) \times 100$$

- **Input Weight:** Sum raw weight fed directly into operational factory hardware lines.
- **Output Weight:** Total validated product cleared for downstream commercial use.
- **Wastage Correlation:** Drop-offs in performance percentages track direct extraction loss, material discard, or moisture evaporation.

C. Yield & Resource Projections (Forecasting)

The analytical dashboard evaluates sequence data from historical runs to forecast incoming manufacturing behaviors:

- **Linear Trend Projection:** The application evaluates ordered run indexes ($X = 0, 1, 2\ldots$) against measured historical performance percentages (Y). It outputs a standard linear regression trend ($y = mx + c$) utilizing least-squares methodology:

$$m = [\ N\ \Sigma(XY) - \Sigma X\ \Sigma Y\ ] / [\ N\ \Sigma(X^2) - (\Sigma X)^2\ ]$$

$$c = [\ \Sigma Y - m\ \Sigma X\ ]\ /\ N$$

The trend line models incoming run efficiency expectations. Positive operational trajectory predictions scale subsequent material yield estimates upward.

- **Moving Average Projection:** Tracks the localized mathematical average across the preceding 5 completions to serve as an immediate performance baseline.
- **Resource Utility Projections:** Generates real-time fuel and electricity metrics per physical mass unit based on historical profiles:

$$\text{Firewood Rate} = \text{Total Firewood Consumed} \ /\ \text{Total Input Weight}$$

$$\text{Electricity Rate} = \text{Total Electricity Units Consumed} \ /\ \text{Total Input Weight}$$

The forecasting tool multiplies these baseline consumption coefficients by the user's upcoming volume goals to predict operational resource needs.

4. Unique Key & Code Generation Formulas

The core platform enforces complete systemic uniqueness across item directories and production runs via automated algorithmic naming conventions.

A. Parts / Product Unique Code

Assembles individual product identities automatically upon structural group selection using date-stamped information structures.

$$\text{Format: } P - [\text{CategoryShortCode}] - [\text{YearShort}][\text{JulianDayOfYear}] - [\text{SequenceNo}]$$

- **P-** : Standard static database system indicator specifying an asset catalog entry.
- **[CategoryShortCode]** : A fixed 3-letter capital identifier representing product classification (e.g., Mechanical = `MAC`, Electrical = `ELE`, Raw Materials = `RAW`).
- **[YearShort]** : The trailing 2 digits tracking the active registration calendar year (e.g., year 2026 logs as `26`).
- **[JulianDayOfYear]** : Three-digit sequential tracking code mapping standard year values from `001` to `365/366` (e.g., June 4th acts as day `155`).
- **[SequenceNo]** : A dual-digit auto-incrementing transactional index that starts at `01` and drops back to zero daily per category block.

> **Example Asset Identification Code:** `P-MAC-26155-02` (Represents the second mechanical component registered in the system on June 4, 2026).

B. Production Batch Code

Assembles a tracking code on manufacturing lines to trace output batches back to raw material suppliers.

$$\textit{Format: [SupplierShortCode]} - \textit{ALE[YearShort][JulianDayOfYear]}$$

- **[SupplierShortCode]** : Unique text marker mapping the primary source material partner (e.g., `CHAMINDA`).
- **ALE** : Static structural token marking the centralized factory batch execution engine.
- **[YearShort] & [JulianDayOfYear]** : Standard short calendar year and 3-digit padded Julian calendar calculation tracking the processing initialization timestamp.

> **Example Batch Execution Code:** `CHAMINDA-ALE26002` (Traces raw inventory supplied by partner Chaminda entering production lines on the second day of the year 2026).

C. Standard Sequence Keys

Generic platform documents use structured, sequential key indexes:

Document Workspace Module	Standard Key Structure	System Operational Example
Products (Fallback Default)	`PRD-XXXX`	PRD-1001
Bills of Materials (BOM)	`BOM-XXXX`	BOM-1002
Purchase Orders (PO)	`PO-XXXX`	PO-1045
Sales Orders (SO)	`SO-XXXX`	SO-2011

5. System Security & Operating Guidelines

Concurrency and Race-Condition Protection

To preserve data integrity when multiple operators enter entries or confirm material receipts at identical milliseconds:

- The system uses low-level atomic updates via MongoDB `findOneAndUpdate` instructions within the Mongoose layer.
- Instead of performing separate read and write phases, the infrastructure uses a single atomic `$inc` modifier directly inside database memory. The transaction engine handles these edits in sequence, ensuring that unique counter keys are generated cleanly for each user.

- Unique indexing constraints are applied to key database attributes (`productCode`, `batchNo`). If an unexpected network event bypasses frontend application protections, the core database layer blocks duplicates to prevent data corruption.

Server Timezone Configurations

Julian date metrics convert through fixed UTC calculations. This standard maintains formatting tracking uniformity whether clients or web nodes initiate transactions via regional timezones (e.g., GMT+5:30) or absolute global baselines.

> **Important Operator Policy:** Do not change database server local time configurations manually. Unauthorized changes can break Julian date calculation offsets.

Data Retention and Cleanup

Tracking indices inside the `PartCounter` database schema feature a rolling **30-day Time-To-Live (TTL) index**. Automated routines clear historical counter files after 30 days of inactivity, keeping the storage infrastructure optimized without modifying historical records.

System Administration Support Dashboard Credentials

Administrative Username: `superadmin@example.com` • Administrative Master Key: `Admin123!`